

Table 2: Bray-Curtis dissimilarity, PERMANOVA results

	Df	SumOfSqs	R2	F	Pr(>F)
Year	2	1.418300	0.4839117	23.91015	1e-04
Residual	51	1.512607	0.5160883	NA	NA
Total	53	2.930907	1.0000000	NA	NA

Table 3: SIMPER contributions to Bray-Curtis dissimilarity between 2019 and 2022

	p	cumsum	average	sd	ratio	ava	avb
S. siderea	0.0756	0.1346859	0.0347707	0.0158621	2.1920595	7.4902931	4.5764086
A. tenuifolia	1.0000	0.2351638	0.0259396	0.0213503	1.2149513	4.0729789	4.3909286
A. agaricites	1.0000	0.3344230	0.0256249	0.0194952	1.3144183	9.2507256	7.6419089
P. porites	1.0000	0.4043444	0.0180510	0.0136873	1.3188162	4.4969944	4.2148864
P. astreoides	0.9986	0.4708795	0.0171768	0.0132215	1.2991608	6.9125790	6.2560900
Orbicella spp.	0.9966	0.5325797	0.0159286	0.0127039	1.2538392	2.5118612	2.9466024
P. strigosa	0.6795	0.5924961	0.0154681	0.0113586	1.3618026	1.9997115	1.0786081
D. stokesii	0.0003	0.6504652	0.0149654	0.0084288	1.7755065	1.3809112	0.1666667
M. cavernosa	0.9193	0.7031713	0.0136067	0.0085339	1.5944363	2.4368831	1.3817230
S. intersepta	0.9835	0.7520253	0.0126122	0.0100961	1.2492157	2.2381297	1.9525785
M. decactis	0.0461	0.8002444	0.0124483	0.0124778	0.9976370	0.9781269	0.3874829
H. cucullata	0.0073	0.8478398	0.0122873	0.0088843	1.3830344	1.1637915	0.4051405
M. meandrites	0.0001	0.8838160	0.0092877	0.0079698	1.1653562	0.7668399	0.0000000
E. fastigiata	0.0010	0.9126971	0.0074560	0.0070574	1.0564735	0.6422488	0.0555556
D. labyrinthiformis	0.6723	0.9402960	0.0071250	0.0077200	0.9229229	0.4349126	0.4253808
S. radians	0.1525	0.9600638	0.0051033	0.0066338	0.7692894	0.2452341	0.3184473
A. fragilis	0.8791	0.9694853	0.0024323	0.0048365	0.5029003	0.0555556	0.1666667

	p	cumsum	average	sd	ratio	ava	avb
C. natans	0.0745	0.9768699	0.0019064	0.0042789	0.4455424	0.1666667	0.0000000
Mycetophyllia spp.	0.2250	0.9835181	0.0017163	0.0048855	0.3513108	0.1341230	0.0000000
M. auretenra	0.6880	0.9898991	0.0016473	0.0049761	0.3310454	0.0962250	0.0555556
P. clivosa	0.2410	0.9948563	0.0012798	0.0036402	0.3515727	0.1111111	0.0000000
F. fragum	0.5971	0.9975689	0.0007003	0.0028984	0.2416127	0.0555556	0.0000000
Isophyllia spp.	0.9501	1.0000000	0.0006276	0.0025964	0.2417207	0.0555556	0.0000000

Table 4: SIMPER contributions to Bray-Curtis dissimilarity between 2022 and 2024 (up to 70% cumulative sum of contributions)

	p	cumsum	average	sd	ratio	ava	avb
A. agaricites	0.0002	0.2084315	0.0764400	0.0308723	2.4760095	7.6419089	3.2779923
A. tenuifolia	0.0001	0.3852132	0.0648328	0.0366562	1.7686709	4.3909286	0.8245502
P. porites	0.0001	0.5214982	0.0499811	0.0280779	1.7800844	4.2148864	1.4317997
P. astreoides	0.0177	0.6057145	0.0308854	0.0279862	1.1035936	6.2560900	5.1765121
Orbicella spp.	0.0001	0.6898953	0.0308724	0.0233790	1.3205220	2.9466024	1.4549419
S. siderea	0.9999	0.7529542	0.0231262	0.0158406	1.4599334	4.5764086	3.5251888
S. intersepta	0.0219	0.8023921	0.0181308	0.0142227	1.2747772	1.9525785	1.8877000
P. strigosa	0.8234	0.8433832	0.0150331	0.0112614	1.3349161	1.0786081	0.8902544
M. cavernosa	0.6560	0.8842577	0.0149903	0.0112828	1.3285933	1.3817230	1.2911947
D. labyrinthiformis	0.2279	0.9067034	0.0082317	0.0100298	0.8207272	0.4253808	0.1666667
M. decactis	0.9861	0.9251537	0.0067664	0.0127105	0.5323484	0.3874829	0.0555556
H. cucullata	0.9994	0.9429102	0.0065120	0.0107325	0.6067559	0.4051405	0.0000000
D. stokesii	1.0000	0.9586923	0.0057879	0.0086429	0.6696727	0.1666667	0.2452341
S. radians	0.1233	0.9729922	0.0052443	0.0088748	0.5909230	0.3184473	0.0000000
A. fragilis	0.0658	0.9866897	0.0050234	0.0088435	0.5680366	0.1666667	0.1571348

	p	cumsum	average	sd	ratio	ava	avb
Isophyllia spp.	0.4095	0.9923202	0.0020649	0.0058806	0.3511404	0.0000000	0.1111111
M. auretenra	0.6229	0.9969982	0.0017156	0.0050341	0.3407957	0.0555556	0.0555556
E. fastigiata	1.0000	1.0000000	0.0011009	0.0045522	0.2418326	0.0555556	0.0000000
C. natans	0.9662	1.0000000	0.0000000	0.0000000	NaN	0.0000000	0.0000000
P. clivosa	0.8901	1.0000000	0.0000000	0.0000000	NaN	0.0000000	0.0000000
F. fragum	0.6616	1.0000000	0.0000000	0.0000000	NaN	0.0000000	0.0000000
M. meandrites	1.0000	1.0000000	0.0000000	0.0000000	NaN	0.0000000	0.0000000
Mycetophyllia spp.	0.9349	1.0000000	0.0000000	0.0000000	NaN	0.0000000	0.0000000

Table 5: SIMPER contributions to Bray-Curtis dissimilarity between 2019 and 2024 (up to 70% cumulative sum of contributions)

	p	cumsum	average	sd	ratio	ava	avb
A. agaricites	0.0001	0.2013097	0.0878858	0.0274196	3.2052208	9.2507256	3.2779923
S. siderea	0.0001	0.3347599	0.0582604	0.0197924	2.9435797	7.4902931	3.5251888
A. tenuifolia	0.0020	0.4451992	0.0482145	0.0289765	1.6639164	4.0729789	0.8245502
P. porites	0.0001	0.5494477	0.0455118	0.0239539	1.8999766	4.4969944	1.4317997
P. astreoides	0.1161	0.6166778	0.0293507	0.0252265	1.1634836	6.9125790	5.1765121
P. strigosa	0.0011	0.6614561	0.0195489	0.0130038	1.5033210	1.9997115	0.8902544
M. cavernosa	0.0009	0.7061294	0.0195030	0.0127774	1.5263670	2.4368831	1.2911947
Orbicella spp.	0.8278	0.7501664	0.0192252	0.0136843	1.4049081	2.5118612	1.4549419
D. stokesii	0.0001	0.7903099	0.0175255	0.0102091	1.7166459	1.3809112	0.2452341
H. cucullata	0.0001	0.8294216	0.0170750	0.0108284	1.5768715	1.1637915	0.0000000
S. intersepta	0.7771	0.8627205	0.0145373	0.0122244	1.1892023	2.2381297	1.8877000
M. decactis	0.0005	0.8959962	0.0145272	0.0157686	0.9212719	0.9781269	0.0555556
M. meandrites	0.0001	0.9222873	0.0114779	0.0098876	1.1608424	0.7668399	0.0000000

	p	cumsum	average	sd	ratio	ava	avb
E. fastigiata	0.0001	0.9433695	0.0092039	0.0087328	1.0539398	0.6422488	0.0000000
D. labyrinthiformis	0.6721	0.9596516	0.0071083	0.0090811	0.7827534	0.4349126	0.1666667
S. radians	0.8283	0.9677289	0.0035263	0.0067486	0.5225285	0.2452341	0.0000000
A. fragilis	0.7744	0.9742532	0.0028483	0.0067211	0.4237899	0.0555556	0.1571348
C. natans	0.0001	0.9795726	0.0023223	0.0052054	0.4461289	0.1666667	0.0000000
Isophyllia spp.	0.3111	0.9848237	0.0022925	0.0054032	0.4242799	0.0555556	0.1111111
Mycetophyllia spp.	0.0001	0.9897560	0.0021533	0.0061104	0.3523966	0.1341230	0.0000000
M. auretenra	0.4222	0.9944228	0.0020374	0.0061126	0.3333067	0.0962250	0.0555556
P. clivosa	0.0001	0.9980023	0.0015627	0.0044434	0.3516897	0.1111111	0.0000000
F. fragum	0.0001	1.0000000	0.0008721	0.0036046	0.2419550	0.0555556	0.0000000